

Ira R. Check

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SKILLS

- **Languages:** C#, Java, Python, C, C++, HTML, CSS, SQLite
- **Frameworks and Libraries:** .NET, WPF/XAML, JavaFX, Swing, Pytest (Unit Testing), OpenCV
- **Tools:** Apache Tomcat, Git, GitHub, Visual Studio, IntelliJ, CLion
- **Other:** Unity3D

EDUCATION

The Pennsylvania State University

*Bachelor of Science in Computer Science,
Minor in Cybersecurity Computational Foundations*

Anticipated: 2027

University Park, PA

- **GPA:** 3.32
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Systems Programming, Calculus I-III, Discrete Math

SOFTWARE PROJECTS

Modular Agent-Based Simulation Engine

June 2025 - August 2025

C# / .NET; ScriptHookV, ScriptHookVDotNet, LemonUI

- Engineered a high-performance simulation capable of handling 80+ AI agents with only 2ms average CPU time per tick and a memory footprint under 35MB per tick.
- Implemented a modular ruleset system and a custom data parser for user-generated content, improving flexibility and extensibility.
- Released to 200+ unique users, validating real-world performance, usability, stability, and most importantly enjoyability on hundreds of machines.

Accessibility Input Controller Tool

February 2025

C# / .NET; WPF / XAML; XInput, WindowsAPI

- Developed a C#/.NET accessibility tool translating controller inputs into desktop commands using XInput and WindowsAPI, improving ergonomics for users with physical impairments.
- Implemented multithreaded input detection, reducing CPU load by 20% while maintaining low-latency response for mouse and media controls.
- Designed an intuitive WPF/XAML interface for accessible user interaction, with a specific emphasis on responsiveness and usability.

Checkers-playing Robotic Arm

February 2026 - Present

Python, OpenCV, Pyserial; C, C++

- Engineered a complete game engine in Python featuring full rules enforcement, board state management, and minimax AI with alpha-beta pruning.
- Designed command serialization layer to interface with robotic arm hardware, collaborating with mechanical and electrical engineering teammates.
- Using OpenCV to automatically determine physical inputs and verify them with the Python brain.

WORK EXPERIENCE

The Giant Company

May 2024 - September 2025

Associate Center Store Replenishment

- Trained new hires in the Center Store department on store operations, procedures, and quality assurance.
- Assumed responsibility for recording high-turnover inventory stock levels during staff-absence.